

A1lyPreserve: Measuring Accessibility Preservation in DOCX-to-PDF Conversion

Anonymous for review

Abstract

Accessible source documents can contain author-supplied information such as heading structure, alternative text, language metadata, list semantics, table relationships, and footnote associations. When a document is converted, inspecting only the destination does not establish whether those source properties survived. We present A1lyPreserve, a source-grounded method that freezes controlled accessibility fixtures, verifies their DOCX representations, converts identical sources through real pipelines, extracts destination PDF structure, and compares source and destination properties with provenance-traceable evidence. The completed benchmark contains 13 atomic DOCX fixtures and 26 genuine DOCX-to-PDF outputs from LibreOffice and the Google Docs DOCX-to-PDF conversion pipeline. All outputs were parseable, tagged, non-empty, and validated. Across the 26 fixture/pipeline cases, 11 were preserved, 9 partially preserved, 3 altered, 2 lost, and 1 remained a measurement error; the 25 determinate cases therefore contain 11 preserved, 9 partially preserved, 3 altered, and 2 lost outcomes. The two confirmed losses concern inline language and footnote association in the Google Docs pipeline. Results are feature-specific and are not used to rank converters globally. The study demonstrates that source-grounded preservation is measurable and that tagged destination output alone does not fully characterize whether known source accessibility information survived conversion.

Keywords: document conversion, accessibility preservation, DOCX, PDF, tagged PDF, differential testing, alternative text, document structure

1 Introduction

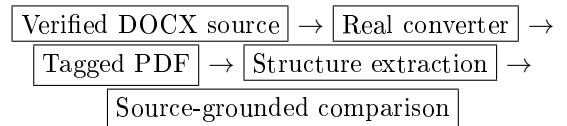
Authors can encode accessibility information directly in source documents: a heading is more than large text, an image may carry an author-written alternative, a run may identify a language change, and a table may identify header relationships. These properties matter because the source document is often exported, uploaded, or converted before it reaches readers. The resulting file may contain some accessibility structure, but that fact alone does not show whether the information intentionally supplied by the author survived the transformation.

This distinction is recognized in accessibility guidance.

ATAG 2.0 addresses preservation during content transformations when equivalent output mechanisms exist [7]. Revised Section 508 includes a requirement to preserve information provided for accessibility during format conversion to the extent supported by the destination format [10]. EN 301 549 likewise specifies preservation of accessibility information in transformations [3]. These requirements motivate a practical question: how can preservation be tested against known source ground truth rather than inferred from the destination alone?

Prior work has studied office-document conversion, PDF-to-HTML accessibility, automated PDF remediation, and destination accessibility evaluation. Roig and Ribera examined office-to-EPUB conversion [8]; SciA1ly and related Paper-to-HTML work reconstruct accessible representations from scientific PDFs [11]; iTagPDF uses source cues to improve PDF tagging [6]; and recent benchmarks evaluate PDF accessibility or conversion services [5, 1, 2]. A1lyPreserve complements these efforts by beginning with controlled DOCX documents whose target accessibility properties are explicitly verified, then measuring whether those properties survive two real DOCX-to-PDF pipelines.

We present A1lyPreserve as a methodology, benchmark, comparison tool, and empirical study. The workflow is:



The frozen benchmark contains 13 atomic DOCX fixtures and 26 genuine PDF outputs: 13 from LibreOffice and 13 from the Google Docs DOCX-to-PDF conversion pipeline. All outputs were parseable, non-empty, tagged, and validated. The frozen outcomes were 11 PRESERVED, 9 PARTIALLY PRESERVED, 3 ALTERED, 2 LOST, and 1 MEASUREMENT ERROR. The two confirmed losses concern inline language and footnote association in the Google Docs pipeline. These findings are not a product ranking and do not estimate the prevalence of conversion problems in arbitrary documents.

The paper makes four scoped contributions:

1. a source-grounded method for measuring whether known accessibility information survives document conversion;

2. a controlled benchmark of 13 verified DOCX accessibility fixtures with explicit source ground truth;
3. `a11ydiff`, an automated, provenance-traceable comparison framework relating source properties to converted PDF structure; and
4. an empirical study of 26 real outputs from two DOCX-to-PDF pipelines showing preservation, partial preservation, alteration, and confirmed loss across tested properties.

The paper deliberately makes no universal claim about accessibility loss during conversion, does not rank either converter globally, and does not treat the results as real-world prevalence. It asks a simpler question: when accessibility information is known to exist before conversion, did that information survive?

2 Background and Related Work

2.1 Preservation as an accessibility requirement

Accessibility information is often authored in one representation and transformed into another. ATAG 2.0 explicitly addresses restructuring and recoding transformations: when equivalent mechanisms exist in the output technology, accessibility information should be preserved or the author should be warned or prompted to check it [7]. Its text-alternative criterion is especially close to the F02 contract. The U.S. Revised Section 508 standards similarly state that authoring tools should preserve information required for accessibility during format conversion to the extent supported by the destination format [10]. EN 301 549 contains a corresponding preservation-of-accessibility-information requirement for transformations [3]. These requirements motivate testing preservation, but they do not predict the outcomes of the converters evaluated here.

2.2 Accessibility-aware document conversion

Roig and Ribera evaluated office-document conversion to EPUB and reported accessibility-preservation problems in that setting [8]. Their work is an important precedent for studying accessibility effects of conversion. A11yPreserve differs in experimental scope and instrumentation: it uses frozen native DOCX source contracts, real DOCX-to-PDF pipelines, destination structure extraction, and provenance-traceable fixture-level differential records. The distinction is one of scope and method, not a claim that prior work ignored preservation.

2.3 Accessible representation generation and remediation

SciA11y converts scientific papers from PDF to accessible HTML and combines extraction with navigation-oriented rendering for blind and low-vision readers [11]. This work primarily creates a more accessible destination representation. iTagPDF uses source-document cues and visual PDF analysis to automate tagging, reading order, and content-specific metadata such as alt text and formula structure [6]. Its source-aware remediation objective is closely related to A11yPreserve’s use of source information, but the experimental question differs: iTagPDF generates or improves PDF metadata, whereas A11yPreserve measures whether author-provided source information survived an existing conversion pipeline.

2.4 PDF conversion and accessibility evaluation benchmarks

Recent work also benchmarks destination conversion or evaluation. Kumar, Padath, and Wang introduce a benchmark for assessing automated and LLM-based PDF accessibility evaluation across criteria including alternative text, reading order, tagging, tables, and hyperlinks [5]. DAISY’s 2026 benchmark compares seven services that convert PDFs into accessible documents against a shared quality framework [1]. DocAccessible’s public research program reports source-grounded PDF-to-HTML conversion and fidelity benchmarking with frozen evidence [2]. These are important adjacent efforts. A11yPreserve’s narrower contribution is to start from controlled DOCX accessibility ground truth and compare the same source semantics across real DOCX-to-PDF pipelines, while separating preservation fidelity from destination-only accessibility evaluation.

The resulting gap is therefore modest: not the invention of accessibility-conversion benchmarking, but a source-grounded DOCX-to-PDF preservation measurement design with explicit contracts, frozen source hashes, feature-level classifications, and a reproducible evidence chain.

3 A11yPreserve Method

A11yPreserve treats accessibility preservation as a source-to-destination comparison problem. The method begins with a document whose target accessibility properties are known, runs that exact source through a named conversion pipeline, extracts accessibility-relevant destination structure, and classifies the correspondence. This framing is intentionally narrower than asking whether a PDF is accessible in every respect. A destination-only checker can report tags or metadata that are useful in their own right, but it cannot by itself establish whether an author-provided source property was retained.

3.1 Source fixtures and ground truth

The frozen corpus contains 13 small DOCX fixtures. The fixtures cover heading hierarchy, image alternative text, list semantics, table/header structure, document language, inline language changes, decorative image state, hyperlinks, document title and metadata, multi-level table headers, footnote association, equation/math semantics, and figure/caption association. They were created by controlled generation scripts using `python-docx` together with explicit `WordprocessingML` constructs where required. Each fixture isolates a target property and avoids unnecessary layout complexity.

Each fixture has a machine-readable manifest. Source ground truth required agreement among three artifacts: the manifest, raw OOXML inspection, and the source extractor’s accessibility intermediate representation (ASIR). OOXML checks inspect the relevant document parts, including document XML, run properties, numbering, relationships, and the footnotes part where applicable. This establishes encoded source ground truth; it does not claim validation by a panel of accessibility experts or by disabled users. The corpus manifest records file size, SHA-256, feature family, and verification status.

3.2 Conversion pipelines

The first condition used LibreOffice 26.2.6.3, build 8221e31b3ac356a1623c672912a3d2b492f7e3d1, on Windows 10.0.26200.0. Conversion used native headless `soffice.com writer_pdf_Export` with `UseTaggedPDF=true`, `PDFUACompliance=true`, and `SelectPdfVersion=1`. The second condition used an authenticated Google Docs web workflow on the same Windows environment with Chrome 153.0.8010.53: upload the exact frozen DOCX and choose *File* → *Download* → *PDF Document (.pdf)*. The observed Google PDF producer was `Skia/PDF m156`. No print-to-PDF path, screenshot, manual content repair, or post-export accessibility repair was used. The official LibreOffice documentation describes its PDF/UA export option [9]; Google’s download documentation describes the selected-file-type download workflow [4].

The Google condition is described throughout as the *Google Docs DOCX-to-PDF conversion pipeline*. It includes DOCX import, Google’s internal document representation, and PDF generation. The experiment does not isolate which internal stage caused an observed change. Microsoft Word was not included in the denominator because it was not a completed primary pipeline.

3.3 Destination extraction and validation

All 26 outputs were checked for existence, non-zero size, parseability, expected page count, structural tagging, and a PDF structure tree. The extractor inspected PDF

structure elements and roles relevant to the fixture contracts: headings, figures and `/Alt`, list containers and `/ListNumbering`, tables and header cells, `/Lang`, link annotations, metadata, `/Note`, `/Formula`, and `/Caption`. Evidence was retained as structured JSON and raw object observations. Poppler `pdftotext` was used as an independent secondary check when marked-content text was not recoverable from the structural parser.

The extractor is deliberately not treated as infallible. For example, the LibreOffice F11 output contains a `/Note` structure, but the available extraction evidence does not establish the note-body association. That case is retained as `MEASUREMENT ERROR` rather than converted into a positive or negative preservation claim.

3.4 Differential comparison

The `a11ydiff` implementation loads the source ASIR and destination extraction for each fixture/pipeline pair. It records the source value, destination value, destination representation, evidence paths, source/output hashes, confidence, and a classification. The feature contracts and classification rules were fixed before the final conversion results were reviewed. The implementation also records a secondary destination-accessibility axis; the primary paper labels are the simpler preservation outcomes below.

PRESERVED The destination expresses the contracted property exactly or through a verified equivalent representation.

PARTIALLY PRESERVED Some required structure or information survives, but a contract component is absent or not independently established.

ALTERED The feature remains present, but an author-relevant value changes, such as alt-text wording, locale specificity, or title metadata.

LOST The source property is valid and representable in the destination, but the required destination representation is absent and independent checks support that absence.

MEASUREMENT ERROR The output is valid and relevant structure is present, but the available evidence cannot resolve the contract reliably.

The method also supports `NOT REPRESENTABLE` and `INVALID CONVERSION` for completeness; neither occurred in the frozen primary result set.

3.5 Independent verification and freezing

Non-preserved cases were reviewed against the manifest, raw OOXML, source ASIR, output validation, primary PDF extraction, raw PDF structure, and independent

text/object checks where appropriate. The two LOST cases were retained only after these checks. Source fixtures and PDF outputs were SHA-256 frozen. Phase 6 independently recomputed all 13 source hashes and 26 output hashes, normalized the primary and `allydiff` classification records, and reconstructed all 26 rows without discrepancy. The future freeze process should add a hash for the test-result report; this documentation gap does not change the frozen experiment.

4 Experimental Design

4.1 Research questions

The experiment addresses three questions:

1. **RQ1:** To what extent do the tested DOCX-to-PDF conversion pipelines preserve accessibility information present in controlled source documents?
2. **RQ2:** Which tested accessibility properties are preserved, partially preserved, altered, or lost during conversion?
3. **RQ3:** How do preservation outcomes differ between the tested LibreOffice and Google Docs conversion pipelines?

Validator comparison is not a fourth core question. The project did not obtain a systematic PAC, Acrobat Accessibility Checker, or veraPDF dataset for all 26 outputs, so destination-only validator comparison is future work rather than a central result.

4.2 Benchmark composition

Table 1 summarizes the 13 atomic fixture contracts. Each fixture contributes one feature-level outcome per pipeline, yielding $13 \times 2 = 26$ cases. The fixtures are intentionally controlled rather than randomly sampled. The table’s contract descriptions state what is compared; they do not imply that the feature families have equal complexity or equal user impact.

4.3 Converter conditions and provenance

The LibreOffice workflow was deterministic and local. The Google Docs workflow was authenticated and browser-mediated, so its service backend was not independently version-pinned. The experiment date, browser version, observed PDF producer, source hashes, output hashes, file sizes, and validation records were retained in the automation log. All outputs were one-page PDFs in this corpus and passed the output-validation checks.

The experiment’s unit is a fixture/pipeline pair, not an independent random observation from a population.

Counts and feature-level observations are therefore primary. No inferential statistics, significance tests, prevalence estimates, or weighted overall converter score are used.

4.4 Outcome accounting

The frozen primary set contains 11 PRESERVED, 9 PARTIALLY PRESERVED, 3 ALTERED, 2 LOST, and 1 MEASUREMENT ERROR case. Thus 26 cases were attempted and successfully converted, while 25 cases have determinate preservation outcomes after excluding the measurement error. When descriptive proportions are shown, the denominator is explicit: among 25 determinate cases, the proportions are 11/25 (44%), 9/25 (36%), 3/25 (12%), and 2/25 (8%), respectively. These values describe this controlled benchmark; they are not estimates of arbitrary-document prevalence.

5 Results

5.1 Overall accounting

All 26 conversion attempts produced genuine, parseable, non-empty, structurally tagged PDFs that passed validation. The primary outcome counts are 11 PRESERVED, 9 PARTIALLY PRESERVED, 3 ALTERED, 2 LOST, and 1 MEASUREMENT ERROR. Excluding the measurement error leaves 25 determinate outcomes: 11 PRESERVED, 9 PARTIALLY PRESERVED, 3 ALTERED, and 2 LOST. The measurement error is not silently assigned to any outcome and is not included in the 25-case outcome percentages.

5.2 Feature-level outcomes

Table 2 is the complete frozen feature-level result matrix. LibreOffice preserved the tested property for F01, F03, F04, F05, F06, F09, F10, F12, and F14; it altered F02, partially preserved F07 and F08, and produced the F11 measurement error. The Google Docs pipeline preserved F02 and F04, partially preserved F01, F03, F07, F08, F10, F12, and F14, altered F05 and F09, and lost F06 and F11. These are feature-specific observations, not a global product ranking.

5.3 Cross-pipeline differences

There are 10 fixture rows where the two pipelines have different frozen primary outcomes. The difference is not a single monotonic pattern. For headings, the LibreOffice sequence matched the source while Google Docs added an H1. For alternative text, Google Docs retained the exact author string while LibreOffice prefixed it. For lists, LibreOffice emitted explicit `/ListNumbering` evidence while Google Docs retained nesting without independently establishing ordered versus unordered type. For inline language and footnotes, the LibreOffice outputs contained relevant structural evidence while the

Table 1: Frozen fixture contracts.

Fixture	Property	Contract focus
F01	Headings	Heading levels and order
F02	Alternative text	Author-written image description
F03	Lists	Ordered/unordered type, nesting, and list structure
F04	Table	Table, row/cell, and header structure
F05	Document language	Primary document language value
F06	Inline language	Run-level language override within an en-US document
F07	Decorative image	Meaningful versus explicitly decorative image state
F08	Hyperlinks	Link existence, visible text, and URI destination
F09	Document title	Visible title and document metadata identity
F10	Complex table	Multi-level headers, spans, and associations
F11	Footnotes	Note body and reference-to-note association
F12	Equation	Native equation representation and visible expression
F14	Caption	Figure/caption text and association

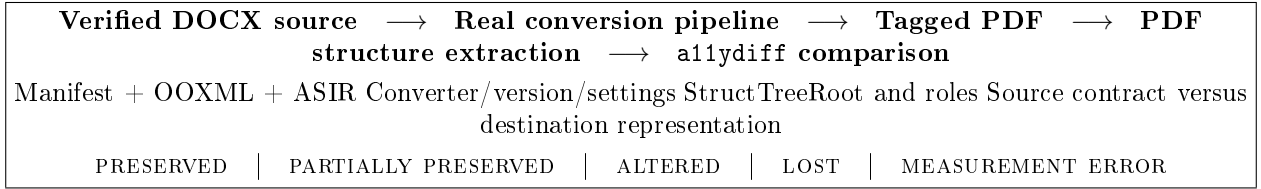


Figure 1: Al1yPreserve’s source-grounded measurement workflow. The destination is inspected, but the outcome is defined by comparison with verified source information.

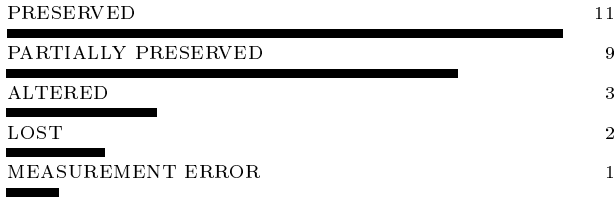


Figure 2: Frozen outcome counts across 26 fixture/pipeline cases. The bars are descriptive counts, not prevalence estimates.

Google Docs outputs did not. These examples show why a feature-level preservation comparison is more informative than a single destination-accessibility verdict.

5.4 Confirmed losses

The strongest negative findings are F06 and F11 under Google Docs. In F06, the source contains the run “Buenos días” with an OOXML **es-MX** override in an **en-US** document. PDF can represent this through a marked span with **/Lang**; the LibreOffice output does so. The Google Docs PDF is valid and tagged, but the destination extractor finds no inline span, raw PDF inspection finds no **es-MX** marker, and **pdftotext** confirms only the visible phrase. The outcome is **LOST** under the pre-specified contract.

In F11, the source contains a native footnote with an explicit reference and note body. The Google Docs PDF is valid and tagged, but raw inspection finds no **/Note**, **/Footnote**, or **/Link** marker, the structural extractor re-

turns no footnote, and independent text extraction does not recover the source note body. Because the PDF structure model can represent the feature and the other pipeline produced a comparable note structure, the result is **LOST**, not **NOT REPRESENTABLE** or **INVALID CONVERSION**.

5.5 Measurement error and altered information

F11 under LibreOffice is deliberately not described as preserved or lost. The output contains a **/Note** structure, but the extractor and independent text check cannot resolve the note-body association. This is a measurement error caused by insufficient destination evidence, not evidence of malformed output.

The three **ALTERED** cases change author-relevant information while retaining a related destination value: LibreOffice prefixes F02 alt text, Google Docs changes F05 from **en-US** to **en**, and Google Docs changes the F09 metadata title. Altered does not automatically mean unusable; it means that exact author-provided information was not retained unchanged.

6 Discussion

6.1 What the benchmark demonstrates

The primary result is methodological: accessibility preservation can be measured when source properties are explicitly known and destination representations can be

Table 2: Complete frozen preservation outcomes.

Fixture and property	Evidence-centered observation	LibreOffice		Google Docs	
F01 headings	Extra H1 changes the Google Docs sequence	PRESERVED		PARTIALLY PRESERVED	PRE-PRESERVED
F02 alt text	LibreOffice changes wording; Google Docs retains the source string	ALTERED			
F03 lists	Google Docs lacks independently established list type	PRESERVED		PARTIALLY PRESERVED	PRE-PRESERVED
F04 table	Table, cells, and header roles retained	PRESERVED		PRESERVED	
F05 document language	Google Docs loses locale specificity ('en-US' to 'en')	PRESERVED		ALTERED	
F06 inline language	'es-MX' span present versus absent	PRESERVED		LOST	
F07 decorative image	Empty-alt representation does not establish explicit artifact state	PARTIALLY PRESERVED	PRE-	PARTIALLY PRESERVED	PRE-
F08 hyperlinks	URI/annotation retained; visible-name association not independently recovered	PARTIALLY PRESERVED	PRE-	PARTIALLY PRESERVED	PRE-
F09 title/metadata	Google Docs changes the metadata title value	PRESERVED		ALTERED	
F10 complex table	Complete scope/span evidence versus incomplete associations	PRESERVED		PARTIALLY PRESERVED	PRE-
F11 footnotes	'/Note' exists but association is unresolved versus absent	MEASUREMENT ERROR		LOST	
F12 equation	'/Formula' structure versus visible text only	PRESERVED		PARTIALLY PRESERVED	PRE-
F14 captions	'/Caption' structure versus adjacency-only association	PRESERVED		PARTIALLY PRESERVED	PRE-

inspected. The 26 cases do not collapse into one outcome. Some properties were preserved, some survived only partially, some changed while remaining present, two were confirmed lost, and one could not be resolved with the available evidence.

This distinction matters because destination accessibility and source-information preservation answer different questions. A tagged PDF can contain headings, figures, or tables and still fail to preserve a particular source property. The F02 case is concrete: LibreOffice retained an alternative description but changed the author-provided string. Conversely, a missing destination structure can be a measurement problem rather than a converter loss, as shown by the LibreOffice F11 case. The comparison therefore complements, rather than replaces, destination-only accessibility evaluation.

6.2 Converter-dependent behavior

The same frozen DOCX source produced different outcomes in 10 of 13 feature rows. The differences are property-specific rather than evidence for a universal winner. Google Docs retained F02 alt text exactly but lost F06 inline language and F11 footnote structure in the tested pipeline. LibreOffice preserved explicit list-type evidence in F03 but altered F02 alt text. Such results support reporting the behavior of a named pipeline under a named configuration, not ranking products globally.

The Google findings should also be attributed carefully. The pipeline includes import and internal representation before PDF download. The data establish end-to-end behavior under the recorded workflow, not an isolated defect in Google's PDF renderer.

6.3 Implications for authoring and conversion

Standards and guidance already recognize preservation as a transformation concern. ATAG 2.0 includes a success criterion for preserving accessibility information during restructuring or recoding transformations when equivalent mechanisms exist, including text alternatives for non-text content [7]. Revised Section 508 states that authoring tools should preserve information required for accessibility to the extent supported by the destination format [10]. EN 301 549 includes a corresponding requirement for preservation during transformations [3]. AllyPreserve operationalizes this concern as a reproducible source/destination comparison for a constrained set of DOCX-to-PDF properties.

For document authors and publishing engineers, the practical implication is not that every export must be rejected or that one tool should always be selected. Rather, checking the destination should be complemented by checking whether critical author-supplied properties survived. The frozen benchmark suggests that this check can be automated for selected structures and that cases requiring human or improved parser review should remain explicitly marked.

6.4 Contribution boundary

AllyPreserve is a methodology, benchmark, comparison tool, and empirical study. `allydiff` is the operational component, not the whole contribution. The work does not attempt to build a general PDF checker, determine user preference, estimate population prevalence, or replace disabled-user evaluation. Its contribution is a concrete measurement chain from verified source information to destination structure and a conservative classification.

7 Threats to Validity

7.1 Controlled corpus and feature units

Atomic fixtures make source ground truth and causal interpretation clearer, but they are synthetic and small. Each feature family has one primary fixture, and feature families differ in complexity and accessibility relevance. Consequently, 26 cases are not IID samples and the outcome counts do not estimate real-world prevalence. The paper reports feature-level results first and gives aggregate counts only as descriptive summaries. Integrated documents, justified variants, and broader source corpora are future work.

7.2 Converter and format coverage

The experiment evaluates DOCX-to-PDF only, using two real pipelines. It does not characterize Microsoft Word, ODT, HTML, EPUB, or other conversion routes. Two independent pipelines are sufficient to demonstrate that behavior can vary across pipelines, but not to characterize the converter ecosystem. Microsoft Word was deferred because the environment did not yield a completed primary workflow; it is excluded rather than treated as a semantic result.

7.3 Cloud reproducibility

Google Docs is a cloud service whose backend version cannot necessarily be pinned. The study records the date, operating system, browser, observed renderer, workflow, source hashes, output hashes, and validation evidence. A rerun can nevertheless differ after a service update. The correct interpretation is therefore a dated Google Docs DOCX-to-PDF pipeline condition, not a timeless property of Google Docs.

7.4 Cross-format representation

Source and destination formats express accessibility information differently. The study uses predefined feature contracts and machine-inspectable mappings, such as Word Heading 2 to PDF /H2, a Word run-language override to PDF /Lang, or a native equation to PDF /Formula. Some mappings are incomplete or parser-sensitive. F07, F08, F10, F12, F14, and especially F11 retain explicit evidence limitations. The measurement-error category prevents such limitations from being silently converted into loss or preservation.

7.5 Parser and validator limitations

The primary destination extractor is a structured parser, supplemented by raw object inspection and Poppler text extraction. Independent checks reduce but do not eliminate parser error. Conventional accessibility validators were not available as a systematic dataset and are not

treated as ground truth. The study therefore does not claim complete PDF accessibility or validator performance.

7.6 User impact

No disabled-user study, screen-reader task study, or perceived-usability study was performed. The results show whether encoded source information was found in the destination representation under the contracts; they do not establish uniform consequences for screen-reader users or other disabled readers. A future user-centered study could connect preservation changes to task impact.

7.7 Interpretation and generalization

An ALTERED description may remain usable, and a PARTIALLY PRESERVED structure may still provide substantial accessibility. Conversely, this study does not determine the practical severity of any individual change. The classifications are evidence-centered preservation labels, not a weighted accessibility score. Claims are limited to the frozen fixtures and recorded pipeline conditions.

8 Conclusion

AllyPreserve demonstrates a practical source-grounded way to measure accessibility-information preservation across document conversion. The study froze 13 controlled DOCX accessibility fixtures, verified their source representations, converted them through LibreOffice and the Google Docs DOCX-to-PDF pipeline, inspected destination PDF structure, and compared the paired representations with retained evidence and hashes.

Across 26 real outputs, the frozen benchmark recorded 11 preserved, 9 partially preserved, 3 altered, 2 lost, and 1 measurement-error case. The two confirmed losses were the inline-language and footnote properties in the Google Docs pipeline. The study also found that a tagged destination can retain accessibility structure while changing or failing to preserve particular source information. This is why destination accessibility and source accessibility preservation should be reported as related but distinct questions.

The conclusions are intentionally bounded. The corpus is controlled and not representative, only two DOCX-to-PDF pipelines were evaluated, Google Docs is cloud-versioned, PDF representation equivalence can be parser-sensitive, and no disabled-user study was performed. Within those limits, the evidence supports the feasibility of source-grounded preservation measurement and shows that preservation behavior is feature-specific and pipeline-dependent. Future work can extend the corpus, add integrated documents and variants, compare additional converters, study validator detection, and connect structural changes to user tasks without changing the

central question: did known accessibility information survive conversion?

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